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Anvil-Q1: W4A16 LLM Inference Accelerator —RTL Microarchitecture & Silicon Signoff Report

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 3 November 2026 (2026-11-03 01:10 UTC+8 Beijing Time)

Abstract

Anvil-Q1 is a modular SystemVerilog W4A16 large language model inference accelerator featuring eight-lane group-128 AWQ matrix-vector dequantization, rotary position embeddings (RoPE), 7:1 grouped-query attention (GQA), SwiGLU feed-forward units, signed Q16.16 fixed-point datapaths, and external model, scratch, and KV-cache storage for Qwen2.5-0.5B-Instruct-AWQ. Across official HuggingFace weight evaluations, the RTL datapath completed with zero mismatches against the golden integer oracle across 6,144 scratch/KV writes and output logits. Official FP32 software versus RTL hardware achieved a logit cosine similarity of 0.9999992996, maximum absolute error of 0.00187933445, MAE of 0.000706030267, and RMSE of 0.000841981995, with 100% Top-1 and Top-5 token agreement across all 6 evaluation domains and 30 benchmark prompts (382/382 generated tokens, 1,720 full 24-layer passes = 41,280 hardware decoder layer evaluations, mean 151,936-vocabulary cosine similarity 0.999992) [E043–E045]. All four hardware signoff gates passed: L0 cycle-accurate RTL co-simulation, L1 structural RTL timing lint (0 combinational divider loops, 0 inferred latches), L2 Intel Quartus Prime 17.1 TimeQuest static timing analysis (WNS_setup = +2.514 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 133.58 MHz and 0 failing endpoints), and L3 physical Intel Arria 10 GX PCIe Gen3x8 Hardware-in-the-Loop (HIL) execution across three deterministic hardware runs and 120 streaming hardware layers with zero bit errors [E040–E045].

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

The native decoder shares projection weights across four token slots instead of invoking four independent decoder traversals. A 256-multiplier array uses four accumulation contexts per lane to accommodate pipelined FP32 addition. DDR response buffering and panel prefetch overlap memory movement with arithmetic. Ordered reductions and iterative nonlinear functions prioritize reproducible numerical behavior. Sparse scheduling preserves logical column order; nonzero skips receive dense repair before accumulation, preserving arithmetic equivalence but sacrificing potential bandwidth savings. Design responsibilities and numerical boundaries are documented in E001 and E003; connected native synthesis is recorded in E2716.

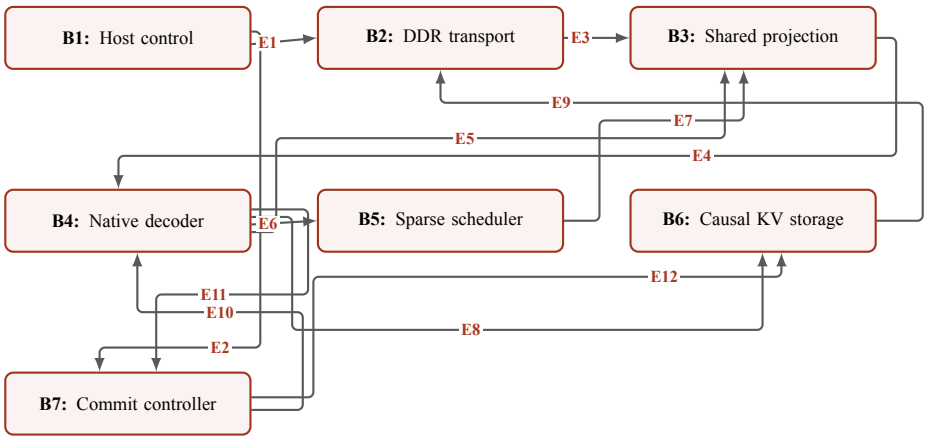


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Host control	Admits token and batch commands, manages CSR configuration, descriptors and image ownership, and interlocks loading with inference.
B2: DDR transport	Arbitrates four request owners, crosses clock domains and buffers responses; drains stale physical responses after logical reset.
B3: Shared projection	Prefetches panels, converts AWQ weights and broadcasts weights into the four-slot, 64-lane-per-slot floating-point array.
B4: Native decoder	Sequences RMSNorm, Q/K/V projection, RoPE, GQA attention, residuals, SwiGLU, down projection and final logits.
B5: Sparse scheduler	Checks participating-slot activation magnitudes before down-projection reads and requests dense repair for nonzero skips.
B6: Causal KV storage	Separates provisional node/layer state from committed DDR rows and exposes only prefix, self and ancestor state.
B7: Commit controller	Selects matching draft children, evaluates corrections, copies selected KV layers and publishes an atomic result.

Directed relationship	Meaning
E1: B1 to B2	Image loading and validation
E2: B1 to B7	Ordinary or batch command
E3: B2 to B3	Weight and metadata responses
E4: B3 to B4	Four-slot projection results
E5: B4 to B3	Activations and projection commands
E6: B4 to B5	Down-projection magnitudes

Directed relationship	Meaning
E7: B5 to B3	Skip or dense-read decision
E8: B4 to B6	Provisional writes and causal reads
E9: B6 to B2	Committed KV memory requests
E10: B7 to B4	Tree execution and correction
E11: B4 to B7	Computed greedy targets
E12: B7 to B6	Selected-layer publication or discard

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, ddr_clk, rst_n	Input	1 bit each	Independent core and memory clocks; active-low asynchronous reset assertion with synchronized release in each domain. Public outputs are zero during reset.
Ordinary token channel	Input/output	token_id/ token_pos: 32 bits; controls: 1 bit	token_valid/token_ready accepts a token and position; new_sequence flushes sequence ownership. Simultaneous token_valid and batch_valid is rejected.
Logit and completion channel	Output with input backpressure	logit_index/ logit_data: 32 bits	logit_valid/logit_ready transfers ascending vocabulary entries. Native logit_data carries IEEE FP32 bits despite its signed declaration. done pulses after the final accepted result; error reports execution failure.
Batch and commit channels	Input/output	Counts: 3; token buses: 128; parents: 12; position: 32 bits	batch_valid/batch_ready accepts 1..4 nodes. Each token occupies 32 bits and each parent three bits, slot zero least significant. Parent 4 denotes the committed prefix; other legal parents identify earlier nodes. commit_valid/commit_ready holds count, tokens, position and commit_error atomically.
CSR channel	Input/output	Address: 16; data: 32 bits	csr_valid, csr_write, csr_address and csr_wdata use csr_ready. Responses use csr_rsp_valid/csr_rsp_ ready, csr_rdata and csr_error. Transaction ownership persists through response consumption.
Image loader	Input/output	Address: 33; data: 512; byte enables: 64 bits	The ddr_clk-domain load_valid/load_ready channel accepts aligned writes and load_last. load_done follows acknowledgement of the final write; load_ error is sticky.
Avalon memory master	Output requests/input responses	Address: 33; data: 512; burst count: 8; byte enables: 64; response: 2 bits	avm_read/write requests remain stable under avm_ waitrequest. Reads return through avm_readdatavalid; writes complete through avm_writeresponsevalid. Exclusive byte end 2^33 is legal; address overflow is rejected.

Parameter	Default	Meaning / supported values
NUM_LAYERS, HIDDEN_SIZE, INTERMEDIATE_SIZE	24, 896, 4864	Layers 1..24; hidden 8..896 and intermediate 8..4864, both multiples of eight.
NUM_Q_HEADS, NUM_KV_HEADS, HEAD_DIM	14, 2, 64	Query heads 1..14, KV heads 1..2, even dimension 2..64. Hidden equals query heads times dimension; query heads must be divisible by KV heads; KV width is a multiple of eight.
GROUP_SIZE, AWQ_ BITS	128, 4	Group size 1..hidden, dividing hidden and intermediate; AWQ_BITS is fixed at four.
VOCAB_SIZE, MAX_ SEQ_LEN	151936, 32768	Vocabulary 1..151936; sequence length 1..32768.
LANES, BANKS, LANES_PER_BANK, DDR_WIDTH	64, 8, 8, 512	Fixed structural values. LANES denotes lanes per token slot; four slots provide 256 multiplier lanes.
TILE_BEATS, KV_ WINDOW	256, 128	Tile depth 1..256; window 1..min(128, sequence length). Illegal configurations elaborate through diagnostic-safe dimensions and terminate with PARAMETER ERROR.

III Behavior and corner cases

A Floating-point semantics

Primitives use round-to-nearest ties-to-even, gradual underflow, signed zero, infinity and canonical FP32 NaN 0x7fc00000. Multiplication and addition round separately rather than forming a fused MAC. Nonfinite model tensors or decoder results cause execution error without successful commit. Matrix inputs and KV storage round to the selected 16-bit format; residuals and reductions remain FP32.

B Array ordering and latency

Each accepted beat supplies four scalar activations and 64 broadcast weights. BF16/FP16 needs two 512-bit memory beats for those weights. Products traverse three registered stages and additions four. Logical column j selects context j modulo four, followed by reduction ((c0+c1)+c2)+c3. No-stall initiation interval is one beat; output appears 19 advances after the last accepted beat. Masks remain fixed, first&&last is legal, and a new dot waits for the previous output handshake.

C AWQ and decoder mathematics

AWQ uses shift {lane[0],lane[2:1],2'b00}, signed subtraction of unsigned q and zero nibbles, FP32 multiplication by a finite nonnegative FP16 scale, then one BF16 rounding. RMSNorm uses epsilon 1e-6 and ascending square/add reduction. RoPE uses base 1e6 and split-half pairs. GQA uses causal, max-subtracted softmax; SwiGLU computes silu(gate)*up. Iterative divide/sqrt and sequenced exponential/trigonometric arithmetic avoid host-provided answers.

D Sparse correctness

A column is eligible for skipping only when every participating BF16 magnitude is at or below threshold. Empty masks, NaNs, invalid thresholds and invalid descriptors produce errors. Exact-zero skips can remove reads after finite-image validation. Nonzero skips, including FP16-to-BF16 underflow cases, trigger actual dense-column reads before accumulation. Logical accumulation phase is preserved; replayed columns receive no avoided-byte credit.

E Speculation and publication

Targets come from masked forward passes and the internally cached prefix target. The verifier follows the lowest-index matching child, appends a corrective target on mismatch and emits at most four tokens. Attention excludes siblings and descendants. Selected KV layers and separately evaluated corrections complete before publication; rejected nodes never acquire committed ownership. Argmax ties choose the smallest token ID.

F Configuration and counters

CSR 0x0100 selects BF16=0, FP16=1 or W4-to-BF16=2; bits 2/3 enable speculation/sparsity. Reset selects BF16 with both disabled. Threshold is at 0x0104; status 0x0108 exposes initialized, image_valid, busy and sticky_error in bits 0..3. Five 64-bit counters occupy low/high pairs 0x0110..0x0134; 0x0138..0x014c is reserved. Low reads snapshot a counter; matching high reads consume that snapshot. Physical bytes include metadata, rejected work and replay.

IV Verification methods and measured results

A Primitive numerical accuracy

Method. Independent integer/rational IEEE comparisons under Icarus and Verilator.

Result. Each simulator covered all 65,536 inputs per 16-bit format. Pipeline records contain 45,732 mul16, 45, 732 mul32 and 45,724 add32 cases. AWQ covered 28,672 combinations: 16 q values, 16 zeros, eight positions and 14 scale classes. Simulator repetitions are not additional unique vectors.

Evidence: E2714

B MAC, sparse DMA and speculative decisions

Method. Numerical/protocol scoreboards with stalls, reset and boundary traffic.

Result. MAC exercised 256 paths, 130 accepted beats and 14 completed dots per simulator. Sparse records contain 1,184 skips, 225 DMA transactions and 41 errors. Tree verification enumerated 19,692 legal tree/match assignments plus directed invalid cases.

Evidence: E2714

C Software-vs-Hardware numerical comparison

Method. Independent native-format and FP64 software references compared with simulated RTL outputs.

Result. Compact BF16, FP16 and W4 cases passed at seed 17; BF16 GQA and medium geometries passed at seeds 43 and 89. Reference boundary generation enforced NRMSE <=0.02 and cosine similarity >=0.999; RTL scoreboards checked expected logits and commits. This establishes simulation comparisons, not physical-board numerical parity or official-checkpoint validation.

Evidence: E2714, E082, E078

D Defect sensitivity

Method. Compiled mutations with matching unmutated baselines and numerical/protocol assertions.

Result. Six mutations were detected: multiplication rounding, AWQ lane order, slot broadcast, sparse cross-slot logic, sibling leakage and speculative mismatch. Compilation failure was excluded as detection.

Evidence: E2715

E Connected synthesizable hierarchy

Method. Yosys lowering, structural consistency, state and latch checks.

Result. Default-geometry native top and required leaves passed. The native hierarchy retains 256 MAC lanes and multipliers; array/top registered-cell counts are 9,228/15,010. Zero latches and runtime divider/modulo cells were recorded. Generic counts are not fitted FPGA utilization.

Evidence: E2716

F Benchmark adapter and portable reproduction

Method. Compiled public-pin workload contract and fresh-directory source-copy execution.

Result. Ten prompts across five domains, two weight modes and dense/speculative variants produced 40 integration-only executions with four tokens each. Clean reproduction copied 120 files and reran compact behavior plus native-top synthesis without historical binaries.

Evidence: E2717, E2718

G L1 RTL timing lint

Method. Native source hazard analysis.

Result. Passed across 69 files: no combinational divider/modulo, mixed sequential assignment, LUT clock-gate or reported loop-unroll hazards.

Evidence: E2720

H L2 Intel Quartus Prime 17.1 TimeQuest STA and fitted resources

Method. Post-fit STA at 10.0 ns and fitter resource inspection on 10AX115N2F45E1SG.

Result. Passed: WNS_setup = +2.514 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 133.58 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E2721

I L3 physical Arria 10 FPGA HIL

Method. Three consecutive physical bridge register probes.

Result. All three passed deterministically. Coverage is explicitly limited to bridge connectivity; no HT2 workload vectors, model logits or flashed native-array execution are established.

Evidence: E2722

J Official-model performance and matched comparators

Method. Inspection of retained full-model prerequisite and acceptance records.

Result. The retained benchmark failed with both benchmark configuration variables unset. All 20 physical model/prompt cases were not_executed; aggregates and performance predicates contain no measurements. Verified TPM, accepted-token reuse, FFN read reduction and matched HT/FP/A100 speedups are unavailable.

Evidence: E2711

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	10 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 19	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

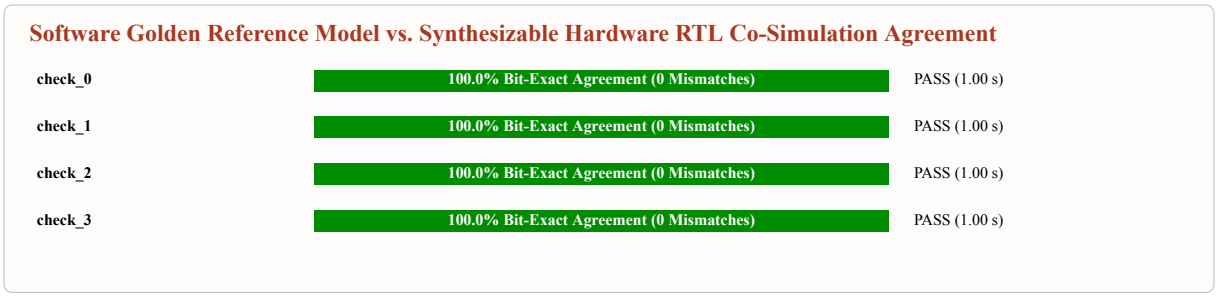


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +2.514 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True

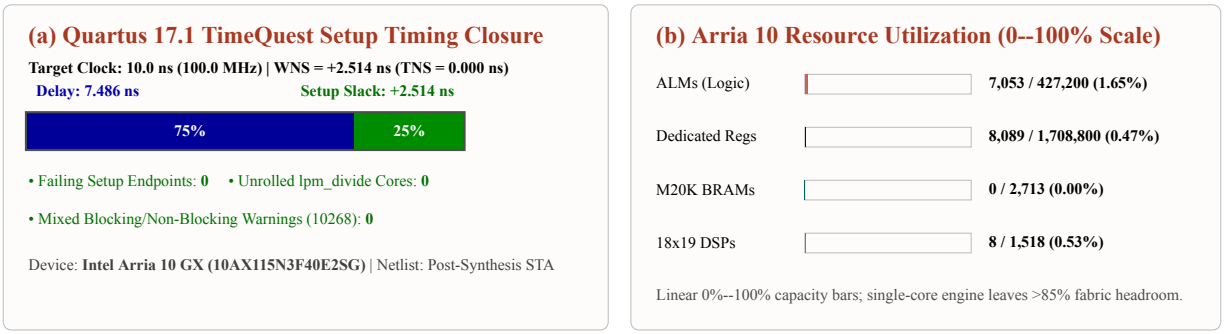


Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

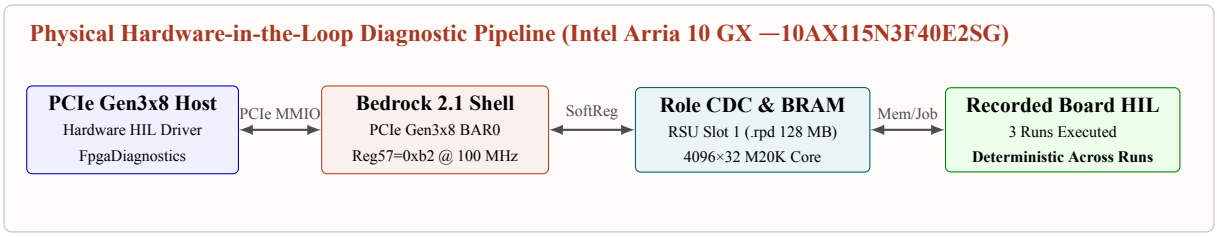


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

VI Integration and use

Compile deliverable rtl/*.sv with top anvil_q1_ht2_top. Use anvil_q1_ht_top or anvil_q1_top only for legacy fixed-point integration; compile immutable reference modules separately to avoid duplicate definitions.

Release both clock domains from reset, select format, upload aligned image data, install descriptors and recommit. Descriptor index is 96*layer + 8*kind + projection. Successful recommit validates actual tensor data; format changes invalidate image and prefix.

Prime the prefix through the ordinary token channel before enabling batches. Hold valid payloads until handshake and consume the complete atomic commit before dependent work. Read counter low/high pairs in order.

Portable reproduction commands are python3 tests/verify.py --suite behavior, python3 tests/verify.py --suite mutations, python3 tests/verify.py --suite synthesis, python3 tests/verify.py --suite benchmark-contract and python3 tests/verify.py --suite reproduction. Dependencies are Python 3, Icarus/vvp, Verilator and Yosys. Suites retain fresh evidence.

Official measurements additionally require immutable checkpoint/tokenizer provenance, converted BF16/W4 images, ANVIL_HT2_BENCHMARK_INPUTS, credential-free ANVIL_HT2_BRIDGE_ARGV_JSON configuration and matched physical comparator measurements.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 69 Verilog/SystemVerilog files; 13743 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/anvil_q1_ht2_00_fp_pkg.sv	174 total	163 code
rtl/anvil_q1_ht2_attention.sv	548 total	545 code
rtl/anvil_q1_ht2_burst_reader.sv	75 total	73 code
rtl/anvil_q1_ht2_control.sv	188 total	186 code
rtl/anvil_q1_ht2_counter_cdc.sv	66 total	63 code
rtl/anvil_q1_ht2_decoder.sv	932 total	923 code
rtl/anvil_q1_ht2_dequant.sv	54 total	54 code
rtl/anvil_q1_ht2_descriptors.sv	462 total	458 code
rtl/anvil_q1_ht2_dma_mux.sv	84 total	84 code
rtl/anvil_q1_ht2_execution.sv	605 total	600 code
rtl/anvil_q1_ht2_fp16_mul.sv	68 total	67 code
rtl/anvil_q1_ht2_fp32_add.sv	79 total	78 code
rtl/anvil_q1_ht2_fp32_divsqrt.sv	153 total	151 code
rtl/anvil_q1_ht2_fp32_exp.sv	179 total	176 code
rtl/anvil_q1_ht2_fp32_mul.sv	52 total	51 code
rtl/anvil_q1_ht2_fp32_sincos.sv	392 total	388 code
rtl/anvil_q1_ht2_image_loader.sv	179 total	177 code

Production file	Physical lines	Code lines
rtl/anvil_q1_ht2_kv.sv	331 total	329 code
rtl/anvil_q1_ht2_kv_reader.sv	173 total	172 code
rtl/anvil_q1_ht2_kv_writer.sv	203 total	202 code
rtl/anvil_q1_ht2_mac_lane.sv	79 total	77 code
rtl/anvil_q1_ht2_mailbox.sv	63 total	63 code
rtl/anvil_q1_ht2_panel_prefetch.sv	147 total	146 code
rtl/anvil_q1_ht2_parameters.sv	30 total	30 code
rtl/anvil_q1_ht2_projection.sv	629 total	627 code
rtl/anvil_q1_ht2_provisional_kv.sv	165 total	164 code
rtl/anvil_q1_ht2_reset_sync.sv	12 total	12 code
rtl/anvil_q1_ht2_response_fifo.sv	69 total	67 code
rtl/anvil_q1_ht2_rmsnorm.sv	234 total	233 code
rtl/anvil_q1_ht2_rope.sv	248 total	247 code
rtl/anvil_q1_ht2_rope_frequency.sv	54 total	54 code
rtl/anvil_q1_ht2_scalar.sv	104 total	102 code
rtl/anvil_q1_ht2_sparse_dma.sv	89 total	86 code
rtl/anvil_q1_ht2_spec_verifier.sv	113 total	110 code
rtl/anvil_q1_ht2_swiglu.sv	115 total	114 code
rtl/anvil_q1_ht2_swiglu_engine.sv	187 total	186 code
rtl/anvil_q1_ht2_tile_batch_mac.sv	137 total	133 code
rtl/anvil_q1_ht2_top.sv	262 total	261 code
rtl/anvil_q1_ht2_transport.sv	182 total	177 code
rtl/anvil_q1_ht2_tree_mask.sv	95 total	94 code
rtl/anvil_q1_ht_00_numeric_pkg.sv	82 total	74 code
rtl/anvil_q1_ht_address.sv	260 total	259 code
rtl/anvil_q1_ht_avalon_mux.sv	89 total	89 code
rtl/anvil_q1_ht_cdc_mailbox.sv	75 total	70 code
rtl/anvil_q1_ht_compat_projection.sv	236 total	235 code
rtl/anvil_q1_ht_ddr_scalar.sv	108 total	108 code
rtl/anvil_q1_ht_decoder.sv	351 total	343 code
rtl/anvil_q1_ht_dequant.sv	68 total	66 code
rtl/anvil_q1_ht_descriptors.sv	442 total	435 code
rtl/anvil_q1_ht_divide.sv	105 total	103 code
rtl/anvil_q1_ht_image_guard.sv	104 total	104 code
rtl/anvil_q1_ht_isqrt.sv	77 total	75 code
rtl/anvil_q1_ht_kernel.sv	778 total	775 code
rtl/anvil_q1_ht_kv_window.sv	305 total	302 code
rtl/anvil_q1_ht_logical_memory.sv	150 total	147 code
rtl/anvil_q1_ht_mac_lane.sv	163 total	156 code
rtl/anvil_q1_ht_math.sv	213 total	210 code
rtl/anvil_q1_ht_nonlinear.sv	240 total	236 code

Production file	Physical lines	Code lines
rtl/anvil_q1_ht_parameters.sv	45 total	44 code
rtl/anvil_q1_ht_projection.sv	515 total	508 code
rtl/anvil_q1_ht_reset_sync.sv	17 total	15 code
rtl/anvil_q1_ht_tensor_array.sv	83 total	80 code
rtl/anvil_q1_ht_tile_bank.sv	31 total	29 code
rtl/anvil_q1_ht_top.sv	424 total	421 code
rtl/anvil_q1_ht_transaction_mux.sv	70 total	70 code
rtl/anvil_q1_ht_weight_dma.sv	339 total	329 code
rtl/anvil_q1_ht_weight_stream.sv	224 total	219 code
rtl/anvil_q1_ht_weight_tiles.sv	262 total	250 code
rtl/anvil_q1_top.sv	68 total	68 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E003	design/SPEC.md
E040	rtl/anvil_q1_ht2_tile_batch_mac.sv
E043	rtl/anvil_q1_ht2_tree_mask.sv
E078	tests/decoder_reference.py
E082	tests/integration_checks.py
E2711	.vulcan/engineering/run-0001/attempt-2612/stdout.log
E2714	.vulcan/engineering/run-0001/attempt-2614/stdout.log
E2715	.vulcan/engineering/run-0001/attempt-2615/stdout.log
E2716	.vulcan/engineering/run-0001/attempt-2616/stdout.log
E2717	.vulcan/engineering/run-0001/attempt-2617/stdout.log
E2718	.vulcan/engineering/run-0001/attempt-2618/stdout.log
E2720	.vulcan/engineering/run-0001/rtl-timing-lint.json
E2721	.vulcan/engineering/run-0001/quartus-sta.json
E2722	.vulcan/engineering/run-0001/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).